

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2021

Bachelor in Computer Applications

Course Title: **Computer Graphics and Animation**

Code No: **CACS305** Semester: **V**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. What do you mean by area filling? Explain boundary fill algorithm.
3. Define scaling transformation? Prove that two successive scaling are multiplicative.
4. What is symmetric property of circle? Digitize a circle whose center is at origin and radius is 9.
5. Define window and view port. Explain 2D viewing transformation pipeline.
6. Reflect a prism A(0,0,0), B(1,1,0), C(1,2,2) and (0,2,0) about yz-plane which has been rotated previously with +90 degree about y-axis.
7. What do you mean by visible surface detection? Explain Z. buffer algorithm for visible surface detection.
8. What is animation? Explain the steps to create animation sequence.

Group C

Attempt any TWO questions[2x10=20]

9. Define refresh rate and persistence of monitor. Explain architecture of random scan display system with block diagram.
10. How can you draw line using Bradenham's line drawing algorithm? Trace it to rasterize the line with end points P(1, 4) and Q(7, 8).
11. Explain Cohen Sutherland line clipping algorithm. Let ABCD be the rectangular window with A(20,20), B(90,20), C(90,70) and D(20,70). Use Cohen Sutherland algorithm to find the region codes and ellipse the lines with end point P(10,30) and Q(80,80)